Analysis of Algorithms

Saghir Ali - 67005

(1) $T(n) = 2T(n/2) + n$

comparing $T(n) = aT(n/b) + f(n)$

- $a = 2$
- $b = 2$
- $f(n) = n$

$n^{\log_b a} = n^{\log_2 2} = n^1 = n$

compare

$f(n) = n$

$n^{\log_b a} = n$

case 2 of master theorem

$$\boxed{T(n) = \Theta(n \log n)}$$

(2) $T(n) = 3T(n/2) + n$

$a = 3$

$b = 2$

$f(n) = n$

$n^{\log_n a} = n^{\log_2 3} \approx n^{1.585}$

$f(n) = n$

$n^{1.585}$

So $f(n) = O(n^{\log_2 3 - \epsilon})$

case 1

$T(n) = \Theta(n^{\log_2 3})$